The Shape of Wisdom, Part I: Layerwise Decision Dynamics in Instruction-Tuned Transformers

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Abstract

When a transformer answers a multiple-choice question, every layer updates the preference scores for the four answer options. We study this layer-by-layer evolution across 3 instruction-tuned models (Qwen 2.5-7B, Llama 3.1-8B, Mistral 7B-v0.3), each evaluated on 3000 MMLU prompts, using *only* cached per-layer option scores—no new inference is performed, and all results derive from a single deterministic run (seed 12345). We define a *soft margin* δ_{soft} that aggregates all incorrect options via temperature-scaled log-sum-exp, eliminating the jagged discontinuities that arise when the hardest competitor switches identity between layers. Using δ_{soft}, we classify each of the 9000 model–prompt pairs into three regimes—*Stable-Correct*, *Stable-Wrong*, and *Unstable*—based on commitment depth and sign-flip count. In a four-dimensional "decision space" whose axes are the option scores, PCA reveals coherent trajectory bundles separated by regime. A flow field computed over normalised depth and soft margin shows convergent dynamics, with commitment concentrating in late layers. Robustness analysis across temperature $\tau \in \{0.5, 1.0, 2.0\}$ and commitment threshold $M \in \{0.5, 0.75, 1.0\}$ confirms that regime proportions shift smoothly rather than discontinuously. All results derive from cached four-option logit vectors. We make no mechanistic claims about internal circuits; causal attribution is deferred to Part II.

1 Introduction

A transformer answering a four-choice question does not make a single decision. At each layer, the model updates the logit scores of four possible answers. By the final layer, one score dominates and the model commits. What happens in between—the geometry of how the model arrives at that commitment—is the subject of this paper.

Prior work studies transformer internals through probing classifiers Belinkov [2022], Hewitt and Manning [2019], causal tracing Meng et al. [2022], and circuit-level analysis Olsson et al. [2022], Conmy et al. [2023]. These methods typically require running the model, injecting perturbations, or training auxiliary networks. We take a complementary approach: given a cache of per-layer logit scores for four answer options, *what can we learn about how a model decides?*

The simplest summary statistic is the *hard margin*: the score of the correct option minus the score of the strongest incorrect option. But the hard margin has a practical problem. When the identity of the strongest incorrect option changes between layers—a *competitor switch*—the hard margin can exhibit sharp jumps that reflect a change in the comparison target, not a change in the model's confidence. Figure 1 makes this concrete: the hard margin trajectory of a single prompt shows jagged discontinuities at every competitor switch, while a *soft margin* that pools all incorrect options smoothly reveals the underlying decision dynamics.

Our answer to the opening question is structured around three ideas. First, we define a soft margin δ_{soft} that replaces the hardest-competitor comparison with a temperature-scaled log-sum-exp over all incorrect options, resolving competitor-switching artifacts (Section 2). Second, we classify model–prompt pairs into three regimes

Table 1: Models studied. All are 7–8 B instruction-tuned transformers with pinned revisions.

Model	Params	Layers	Prompts
Qwen 2.5-7B-Instruct	7.6 B	28	3000
Llama 3.1-8B-Instruct	8.0 B	32	3000
Mistral 7B-v0.3	7.2 B	32	3000

(Stable-Correct, Stable-Wrong, Unstable) based on the trajectory of δ_{soft} across depth, quantifying commitment depth and sign-flip count without overclaiming sharp phase boundaries (Section 3). Third, we construct a "decision space" by treating the four option scores as coordinates, making explicit that this is a four-dimensional proxy space defined by the readout—not the high-dimensional internal representation (Section 3.3).

Contributions.

1. A soft-margin definition that resolves competitor-switching artifacts inherent in hard-max margins (Section 2.3).
2. A regime classification with measurable commitment depth and flip count, replicated across 3 architectures (Section 3).
3. Decision-space visualizations (PCA trajectories, flow fields) revealing convergent structure without model-internal access (Sections 3.3 and 3.4).
4. Robustness analysis showing smooth regime variation under hyperparameter perturbation (Section 4).

All results derive from 276000 cached layerwise score records. No new model inference was performed.

2 Measurement Setup and Definitions

We study four-choice (A/B/C/D) MMLU questions Hendrycks et al. [2021]. At each transformer layer $l \in \{0, \ldots, L{-}1\}$, we extract logit scores for the four answer tokens via a token-bucket readout nostalgebraist [2020] and work exclusively in this four-dimensional subspace. Layer 0 corresponds to the first logged layer (output of the first transformer block), not the raw embedding. We analyse 3 instruction-tuned models (Table 1), each evaluated on 3000 prompts, giving 276000 total layerwise records.

2.1 Score vector

Let $s_A(l), s_B(l), s_C(l), s_D(l)$ denote the four option scores at layer l, and let c denote the correct answer. The score vector at layer l is $\mathbf{v}(l) = [s_A(l),\ s_B(l),\ s_C(l),\ s_D(l)]$. Normalised depth is $d = l/(L{-}1) \in [0, 1]$.

2.2 Competitors

Dynamic competitor. At each layer, the strongest incorrect option: $k_{\text{dyn}}(l) = \arg\max_{j \neq c} \ s_j(l)$, with ties broken alphabetically.

Fixed competitor. The dynamic competitor at the final layer: $k_{\text{fix}} = k_{\text{dyn}}(L{-}1)$.

Soft competitor. A temperature-scaled aggregation of all incorrect options: $\text{softcomp}_\tau(l) = \tau \cdot \text{logsumexp}_{j \neq c}(s_j(l)/\tau)$.

2.3 Margin definitions

Hard dynamic margin. $\delta_{\text{hard,dyn}}(l) = s_c(l) - \max_{j \neq c} \ s_j(l)$.

Soft margin. $\delta_{\text{soft}}(l; \tau) = s_c(l) - \text{softcomp}_\tau(l)$. Throughout this paper, $\delta \equiv \delta_{\text{soft}}(\cdot\,; \tau{=}1.0)$ unless stated otherwise.

Interpretation. δ_{soft} with $\tau = 1$ equals $s_c - \log \sum_{j \neq c} e^{s_j}$. In a four-choice softmax, $\delta_{\text{soft}} = 0$ corresponds to the point where the correct option's probability equals the total probability mass of all three incorrect options combined. We refer to this as the *zero-margin boundary* ($\delta_{\text{soft}} = 0$).

Why soft? The hard margin can exhibit sharp jumps when k_{dyn} changes between layers—a competitor switch (Figure 1A). The soft margin pools all competitors smoothly, producing a trajectory that reflects the correct option's overall advantage. At $\tau \to 0$, δ_{soft} converges to $\delta_{\text{hard,dyn}}$; at large τ, it converges to $s_c - \bar{s}_{\text{incorrect}}$.

2.4 Derived quantities

Competitor switch indicator. $\text{switch}(l) = \mathbb{1}[l \geq 1 \wedge k_{\text{dyn}}(l) \neq k_{\text{dyn}}(l{-}1)]$.

Flip count. The number of sign changes in the sign sequence of δ across layers, dropping exact zeros.

Commitment layer ℓ^*. The smallest l such that $\delta_{\text{signed}}(l') \geq M$ for all $l' \geq l$, where $\delta_{\text{signed}} = \text{sign}(\delta(L{-}1)) \cdot \delta$ and $M = 0.75$ by default. If no such l exists, $\ell^* = \varnothing$.

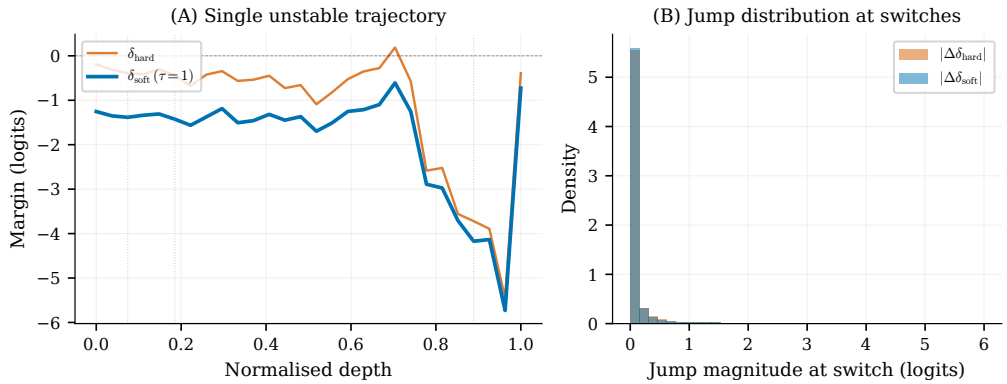

Figure 1: **Why δ_{soft}?** **(A)** Hard margin (vermillion) and soft margin (blue) for the same unstable prompt (Qwen 2.5-7B). Thin vertical lines mark layers where the strongest incorrect option switches identity (competitor switches). The hard margin jumps at each switch; the soft margin tracks smoothly. **(B)** Distribution of jump magnitude at all 64611 competitor-switch events across all models. The hard margin exhibits a heavy right tail; the soft margin dampens switch-induced discontinuities.

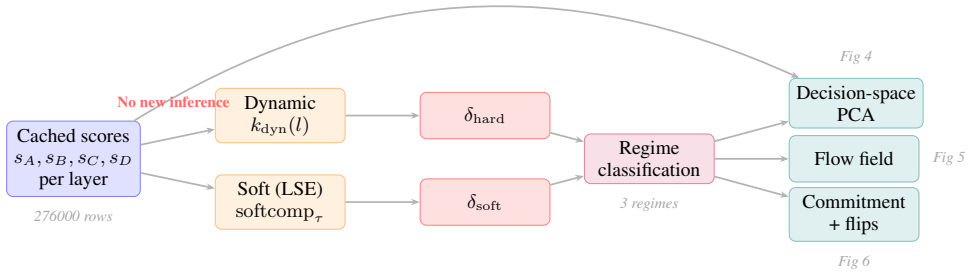

Figure 2: Pipeline overview. Cached per-layer option scores for 3 models flow through competitor definitions and margin computations into regime classification and three downstream analyses. All computations use only the four cached logit scores per layer; no hidden states or weights are accessed.

2.5 Regime classification

A model–prompt pair is classified into one of three regimes. This classification is a **convention**—a useful partition whose boundaries are definitional, not discovered.

- **Stable-Correct.** Flip count ≤ 1, ℓ^* exists, and $\delta(L{-}1) > 0$.
- **Stable-Wrong.** Flip count ≤ 1, ℓ^* exists, and $\delta(L{-}1) < 0$.
- **Unstable.** All other cases.

We explicitly do not claim a natural discontinuity between regimes. Section 4 shows that regime proportions shift smoothly under parameter perturbation.

3 Findings

We analyse 9000 model–prompt pairs (3 models $\times$ 3000 MMLU questions). All figures show $\delta_{\text{soft}}(\tau{=}1.0)$ unless noted.

3.1 Example trajectories

Figure 3 shows representative margin trajectories for each regime. Stable-Correct trajectories rise toward positive values and stay above the commitment threshold $M = 0.75$ for the remainder of depth. Stable-Wrong trajectories mirror this pattern below zero. Unstable trajectories oscillate across the zero-margin boundary, sometimes recovering and sometimes not. Thin dotted lines in stable examples indicate the commitment layer ℓ^*.

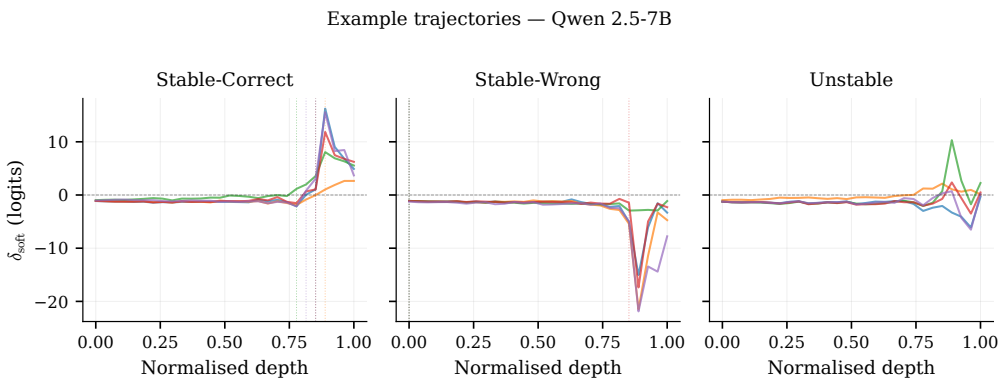

Figure 3: Representative margin trajectories for each regime (Qwen 2.5-7B, five randomly sampled trajectories per regime, seed 12345). Stable-Correct trajectories converge above the zero-margin boundary; Stable-Wrong converge below; Unstable oscillate. Dotted vertical lines mark commitment layers.

Table 2: Regime counts and median final $\delta_{\text{soft}}(\tau{=}1)$ per model.

Model	Regime	N	%	Med. δ
Qwen 2.5-7B	Stable-Correct	1,590	53.0	+5.05
Qwen 2.5-7B	Stable-Wrong	528	17.6	-2.00
Qwen 2.5-7B	Unstable	882	29.4	-0.32
Llama 3.1-8B	Stable-Correct	1,045	34.8	+3.62
Llama 3.1-8B	Stable-Wrong	1,017	33.9	-1.47
Llama 3.1-8B	Unstable	938	31.3	+0.00
Mistral 7B-v0.3	Stable-Correct	1,423	47.4	+5.66
Mistral 7B-v0.3	Stable-Wrong	1,247	41.6	-3.75
Mistral 7B-v0.3	Unstable	330	11.0	-0.09

3.2 Regime proportions and margin distributions

Table 2 reports regime counts per model. Stable-Correct is the majority regime for Qwen (53.0%) but not for Llama (34.8%) or Mistral (47.4%). Stable-Wrong accounts for 17.6% (Qwen), 33.9% (Llama), and 41.6% (Mistral). The Unstable fraction ranges from 11.0% (Mistral) to 31.3% (Llama).

Figure 4 presents the distribution of δ_{soft} at five depth slices across all three models. At the first logged layer (layer 0), all regimes overlap near zero. By normalised depth 0.50, Stable-Correct and Stable-Wrong distributions have begun to diverge. By the final layer (depth 1.0), the three regime distributions occupy largely distinct regions—a consequence of the regime definition, which uses the final-layer sign, combined with the commitment constraint that forces sustained margin above M.

3.3 Decision-space trajectories

The four option scores define a four-dimensional space we call "decision space." We emphasise: this is *not* the model's internal representation space. It is a low-dimensional proxy defined by the four-token readout.

Figure 5 shows PCA projections of $\mathbf{v}(l)$ for each model. Trajectories (paths through layers, shown with depth-graded transparency) form coherent bundles by regime. Final-layer points cluster by regime in the first two principal components, which together explain 92.6% (Qwen), 98.7% (Llama), and 96.8% (Mistral) of variance—unsurprisingly high for a four-dimensional space where the first component captures the dominant correct-vs-incorrect axis.

3.4 Flow field

To visualise aggregate dynamics, we bin the (normalised depth, δ_{soft}) plane and compute two quantities per bin (Figure 6). Panel (A) shows population density: where in the depth–margin plane do data points concentrate? Panel (B) shows mean per-layer drift: the average change in δ_{soft} from one layer to the next.

The flow field reveals three zones of behaviour. At early depth, drift is noisy and population density is concentrated near zero—scores are uninformed. At mid-depth, the mean drift field shows divergence: points above $\delta = 0$ drift further positive, and points below drift further negative. At late depth, drift converges toward the signed final value and population density peaks at the extremes. The yellow band marks the region $|\delta| < \epsilon$ ($\epsilon = 0.2$) where decisions are most

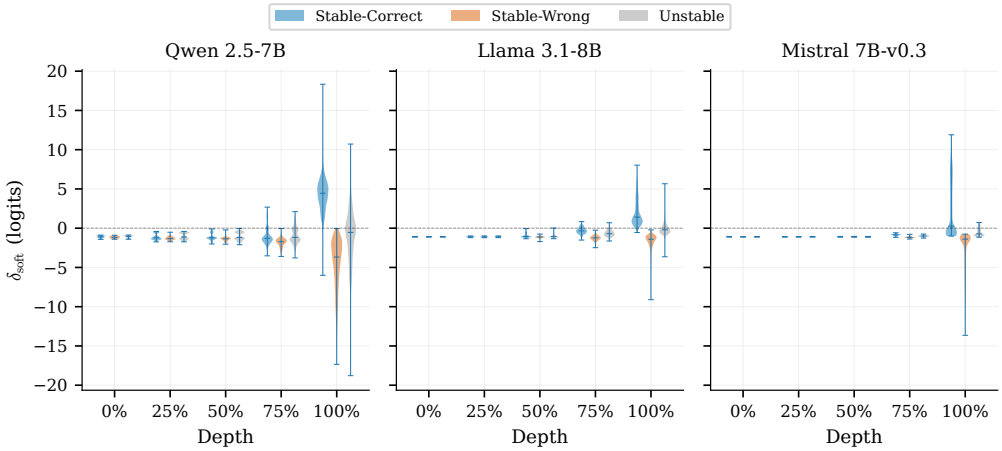

Figure 4: Regime separation emerges across depth. Violin plots of $\delta_{\text{soft}}(\tau{=}1.0)$ at five normalised-depth slices (0%, 25%, 50%, 75%, 100%) for each model, coloured by regime. At depth 0, regimes overlap; by depth 1, they are well separated. Final-layer separation is partly tautological (regimes are defined by final-layer sign); the informative observation is the *gradual* separation at intermediate depths.

fragile. Bins with fewer than 5 observations are masked.

3.5 Commitment and instability

Figure 7 quantifies four aspects of decision dynamics. **(A)** Commitment depth concentrates in late layers: median normalised commitment depth is 0.85 (Qwen), 0.94 (Llama), and 1.00 (Mistral). **(B)** Flip counts follow a heavy-tailed distribution with mean flip counts of 1.20 (Qwen), 0.77 (Llama), 0.52 (Mistral). **(C)** For Unstable prompts, the last sign flip concentrates in the final third of depth, suggesting late-layer processing as the primary source of instability. **(D)** The competitor switch rate peaks at mid-depth and declines toward the output, consistent with the model locking onto a competitor.

4 Robustness

We verify that the regime structure is not an artifact of our particular parameter choices by varying both the temperature τ and the commitment threshold M.

Figure 8 shows regime proportions under $\tau \in \{0.5, 1.0, 2.0\}$ and $M \in \{0.5, 0.75, 1.0\}$, separately for each model. Two patterns emerge. First, higher τ (softer margins) produces slightly fewer Stable classifications as the margin signal is diluted. Second, higher M (stricter commitment) shifts prompts from Stable to Unstable, as expected. Crucially, the transitions are smooth: no parameter combination produces a discontinuous jump in regime proportions. This smooth variation is what we would expect from a continuous underlying distribution rather than a sharp phase boundary.

5 Limitations and Non-Claims

1. **Restricted four-choice readout.** All analyses operate on four option-token logit scores. The model's full vocabulary and internal representations remain invisible.
2. **No mechanistic claims.** This paper deliberately avoids causal claims about attention heads, MLP sublayers, or any internal circuit. We report phenomenology: what the output trajectory looks like, not *why* it looks that way. Causal attribution (targeted intervention experiments, span ablation) is reserved for Part II.
3. **Decision space is a proxy.** The PCA trajectories in Figure 5 live in a four-dimensional option-score space, not in the model's high-dimensional internal representation space. This space captures the readout signal but not the full representational dynamics.
4. **Regime boundaries are conventions.** $\tau = 1.0$, $M = 0.75$, and max flips $= 1$ *define* regime boundaries rather than discover them. Section 4 shows

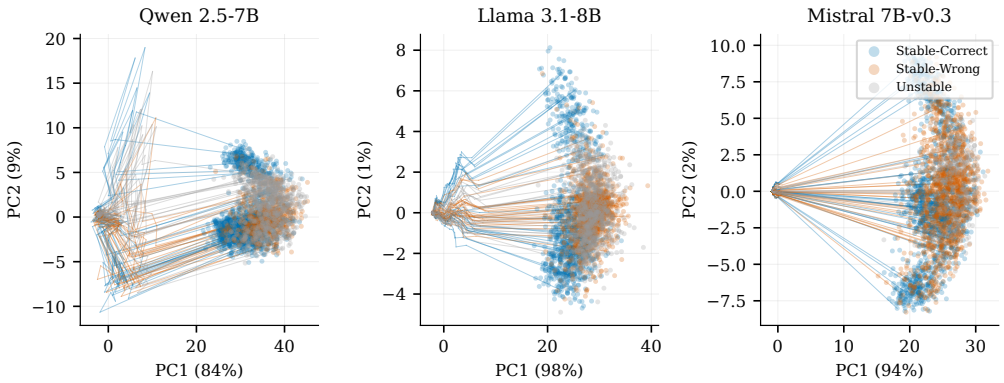

Figure 5: Trajectories in decision space form regime-coherent bundles. PCA of the four-option score vector $[s_A, s_B, s_C, s_D]$ across all layers, one panel per model. Lines trace sampled trajectories with increasing opacity as depth increases; dots mark final-layer positions. PC1 explains 83.7% (Qwen), 97.7% (Llama), 94.4% (Mistral) of variance. This is a proxy *decision* space (4 option scores), not the internal representation space.

smooth degradation, but the partition is a modelling choice.

5. **Scale.** Only 7–8 B models are studied. Scaling behavior at 13 B or 70 B is unknown.
6. **Format.** Only four-choice MMLU; open-ended generation may differ.
7. **Single deterministic run.** All results from seed 12345.

6 Conclusion

A transformer's decision on a multiple-choice question is not a point but a path. By tracking four option scores across every layer, we defined a soft margin that smooths competitor-switching artifacts, classified trajectories into three regimes, and visualised the resulting dynamics in a decision space defined entirely by cached logit vectors.

Three observations stand out. First, regime separation emerges gradually across depth: all regimes overlap at the first logged layer and separate by the final layer. Second, commitment is a late phenomenon, with median normalised commitment depth ranging from 1.00 (Mistral) to 0.94 (Llama). Third, the regime structure is robust to hyperparameter perturbation, varying smoothly rather than discontinuously.

These results establish the phenomenology. What *causes* the observed trajectories—which components push toward or away from the correct answer—is the subject of Part II, which will require additional inference.

Impact Statement

This paper presents an observational analysis of existing model outputs. It does not train models, generate text for end users, or recommend deployment decisions. We do not foresee specific negative societal consequences beyond those generally associated with advances in understanding language model internals.

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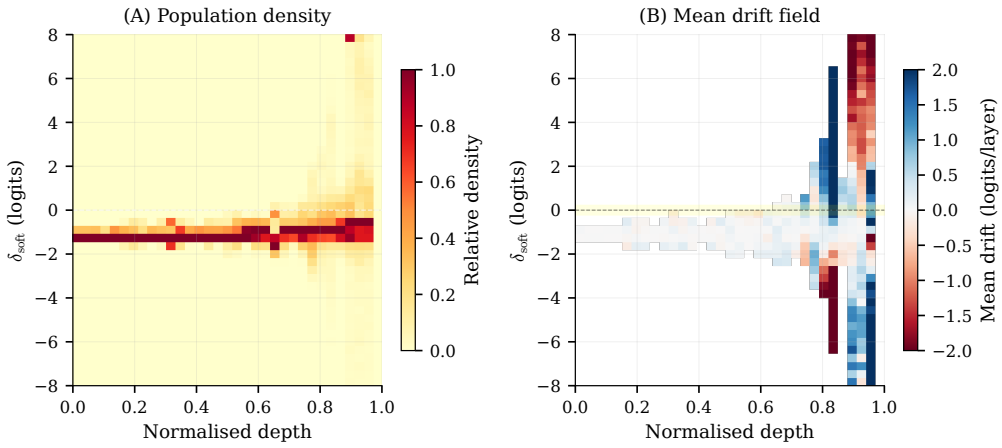

Figure 6: Population density and mean drift in the depth–margin plane, pooled across all 3 models. **(A)** Relative density per depth column: most observations begin near $\delta = 0$ and fan out. **(B)** Mean per-layer drift: blue means positive drift (toward correct), red means negative. Dashed line is the zero-margin boundary ($\delta_{\text{soft}} = 0$); yellow band marks $|\delta| < 0.2$. Bins with count < 5 masked.

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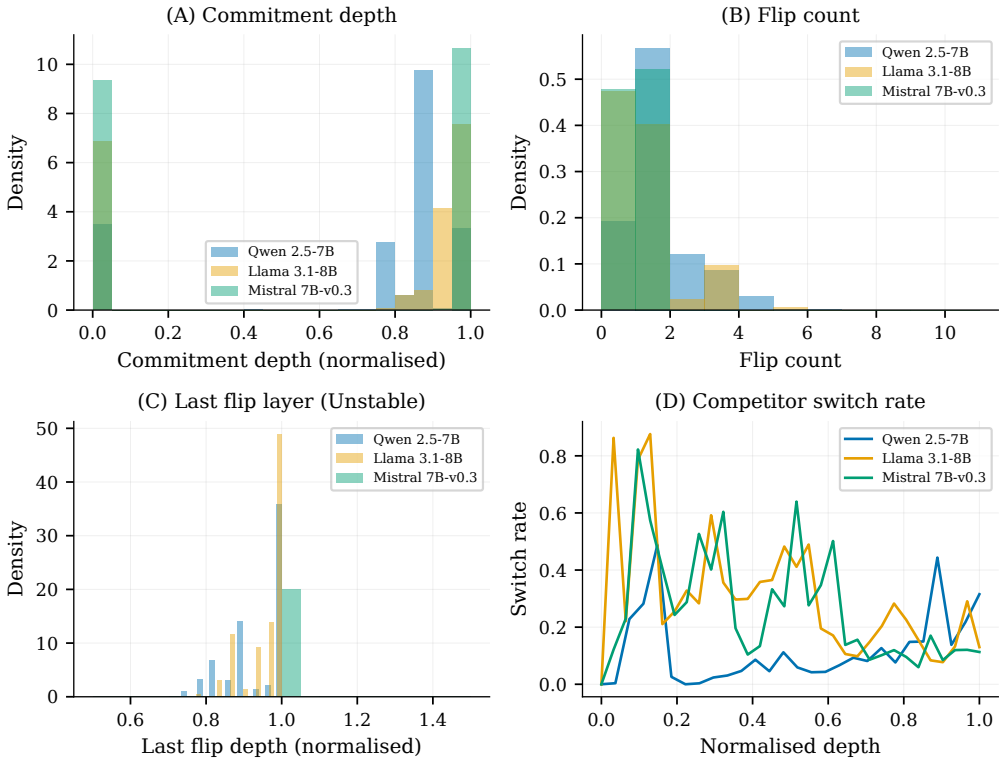

Figure 7: Commitment concentrates late; instability peaks at mid-to-late depth. **(A)** Commitment depth distribution (normalised). **(B)** Flip count histogram. **(C)** Last flip layer for Unstable prompts. **(D)** Competitor switch rate vs. normalised depth. All 3 models shown; $\delta = \delta_{\text{soft}}(\tau{=}1.0)$, $M = 0.75$.

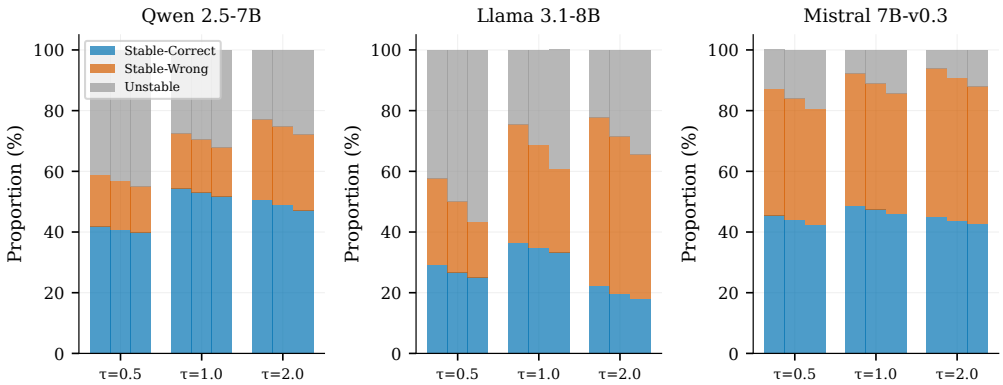

Figure 8: Regime proportions vary smoothly under hyperparameter perturbation. Stacked bars for each model under $\tau \in \{0.5, 1.0, 2.0\}$, grouped by $M \in \{0.5, 0.75, 1.0\}$. No discontinuous jumps appear. Stricter thresholds increase the Unstable fraction; softer margins dilute the Stable fractions slightly.

A Reproducibility

The data, code, and build process are fully specified. The build script `build_part1_assets.py` reads cached parquet files, computes all metrics, generates all figures, and writes `BUILD_INFO.json` containing:

- SHA-256 hashes of every source parquet file.
- SHA-256 hashes of every generated figure PDF.
- Git commit hash and UTC build timestamp.

To reproduce:

```
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 \
  python3 scripts/part1/build_part1_assets.py \
    --parquet-dir results/parquet \
    --output-dir paper/part1
```

All random operations use `numpy.random.Generator` with seed 12345. Groupby operations sort keys (`model_id, prompt_uid, layer_index`) before iteration. Every numeric value in this paper is generated by `compute_numbers.py` and inserted via a LaTeX macro from `auto_numbers.tex`.

B Robustness Details

Flip-count threshold. Setting `max_flips` to 0 (stricter) reduces the Stable classes. Setting it to 2 (looser) expands them. The qualitative three-regime structure is preserved in both cases.

Layer 0 semantics. Layer 0 in our cached data corresponds to the output of the first transformer block, not the raw token embedding. All references to "first logged layer" in this paper refer to this post-first-block output.